SAKARYA ÜNİVERSİTESİ
Bilgisayar ve Bilişim Bilimleri Fakültesi
Bilgisayar Mühendisliği Bölümü

BSM 451
INTERNET OF THINGS AND APPLICATIONS

# IoT Business Models

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# Business Process and Model

- A **business process** or **business method** is a collection of related, structured activities or tasks that produce a specific service or product (serve a particular goal) for a particular customer or customers.(wikipedia).
- There are several types of business processes:
  - Management, Operational, Manufacturing,Logistics, etc.
- A **business model** describes the rationale of how an organization creates, delivers, and captures value (Alexander Osterwalder).
- «a **business model** can best be described through
  - **9** basic building blocks that show the logic of how a company intends to make money.
  - These blocks cover the **4** main areas of a business: customers, offer, infrastructure, and financial viability.
  - The business model is like a blueprint for a strategy to be implemented through organizational structures, processes, and systems.»



**The Business Model Canvas, is a strategic management and entrepreneurial tool. It allows you to describe, design, challenge, invent, and pivot your business model.**

# Why to use Business Model?

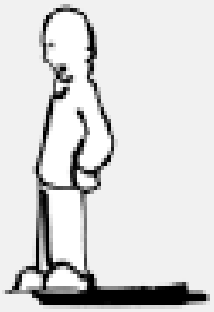
- **Business Model** is used for;
  1. Technology and innovation management,
  2. Strategy
     - Creating Value,
     - Competition edge,
     - Company performance,
  3. e-business
     - How to use information inside the company/organization



**Business model is a bridge between economy and technology.**

**The advancement and changes in technology requires changes in business models.**

# The 9 Building Blocks



CS

**1 Customer Segments**

*An organization serves one or several Customer Segments.*

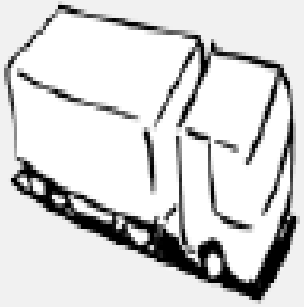


VP

**2 Value Propositions**

*It seeks to solve customer problems and satisfy customer needs with value propositions.*

# The 9 Building Blocks





CH

**3 Channels**

*Value propositions are delivered to customers through communication, distribution, and sales Channels.*

CR

**4 Customer Relationships**

*Customer relationships are established and maintained with each Customer Segment.*

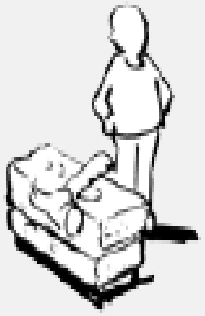
# The 9 Building Blocks



R$

5 **Revenue Streams**

*Revenue streams result from value propositions successfully offered to customers.*



KR

6 **Key Resources**

*Key resources are the assets required to offer and deliver the previously described elements...*

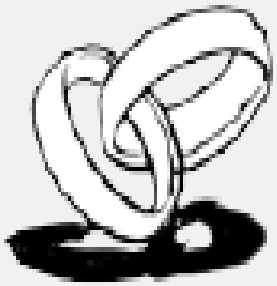


KA

7 **Key Activities**

*...by performing a number of Key Activities.*

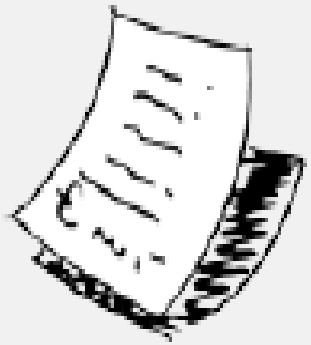
# The 9 Building Blocks



KP

8 **Key Partnerships**

*Some activities are outsourced and some resources are acquired outside the enterprise.*



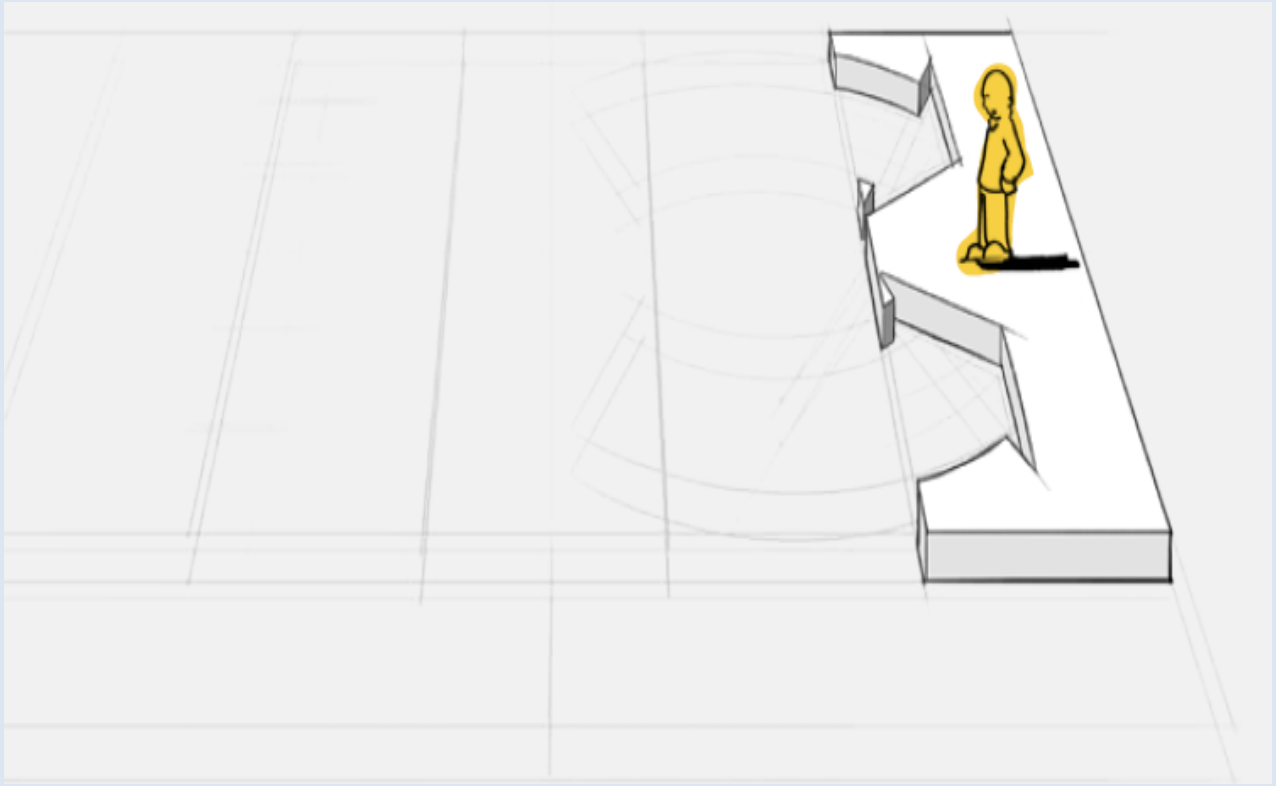
C$

9 **Cost Structure**

*The business model elements result in the cost structure.*

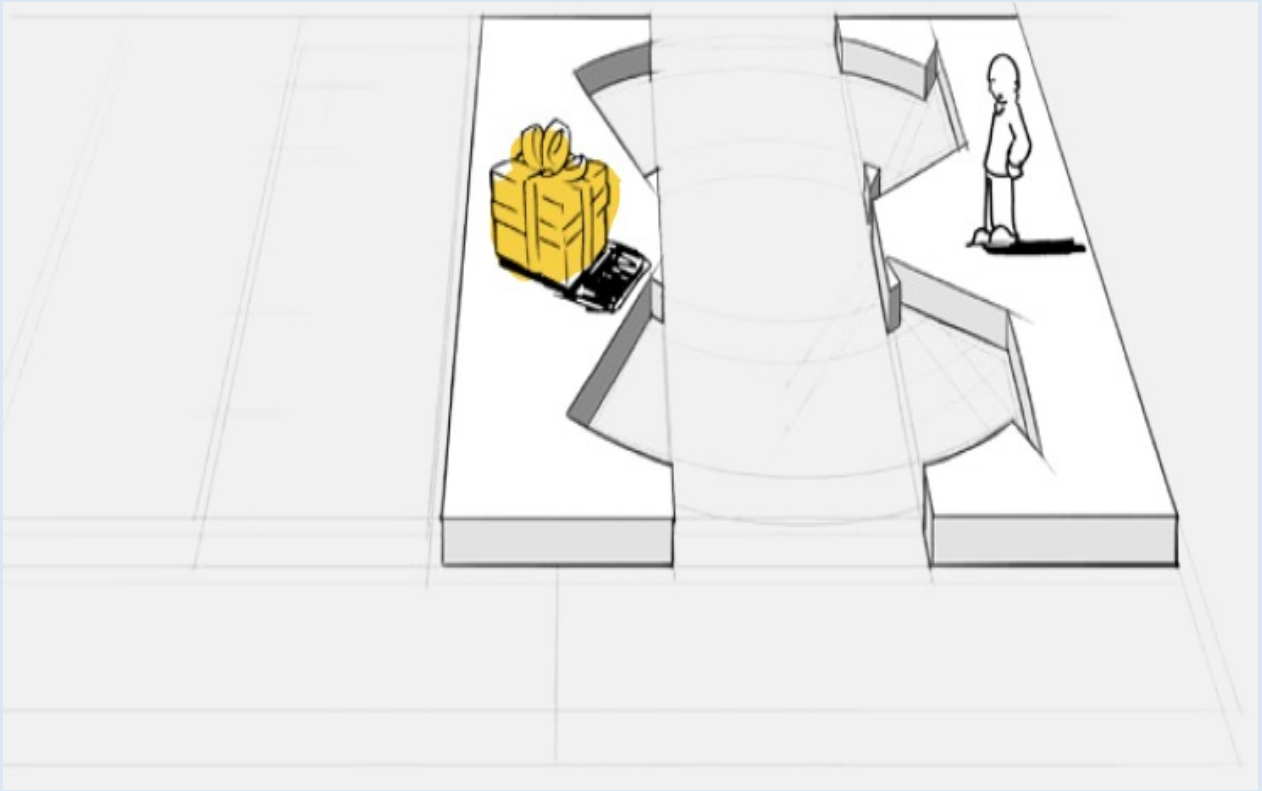
# Customer Segments

**The Customer Segments Building Block** defines the different groups of people or organizations an enterprise aims to reach and serve



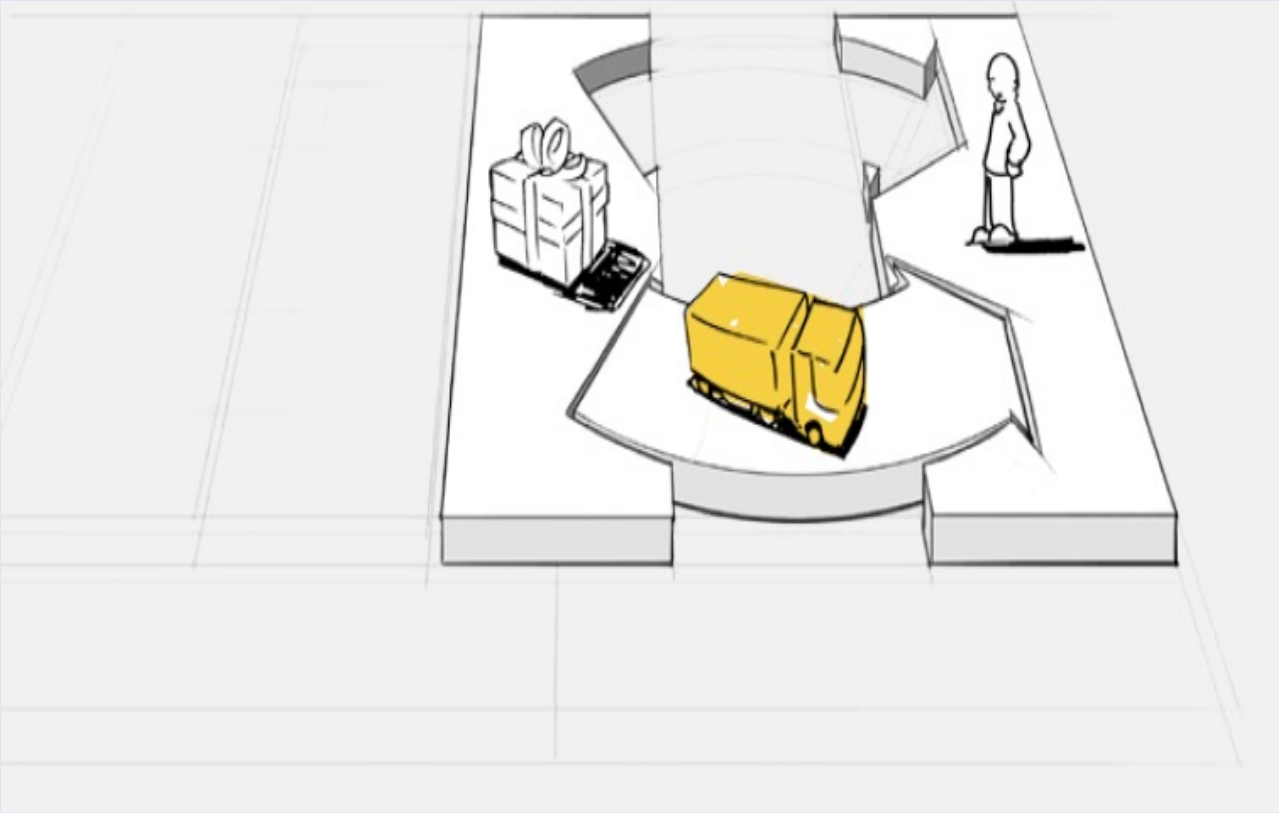
# Value Proposition

**The Value Propositions Building Block** describes the bundle of products and services that create value for a specific Customer Segment.



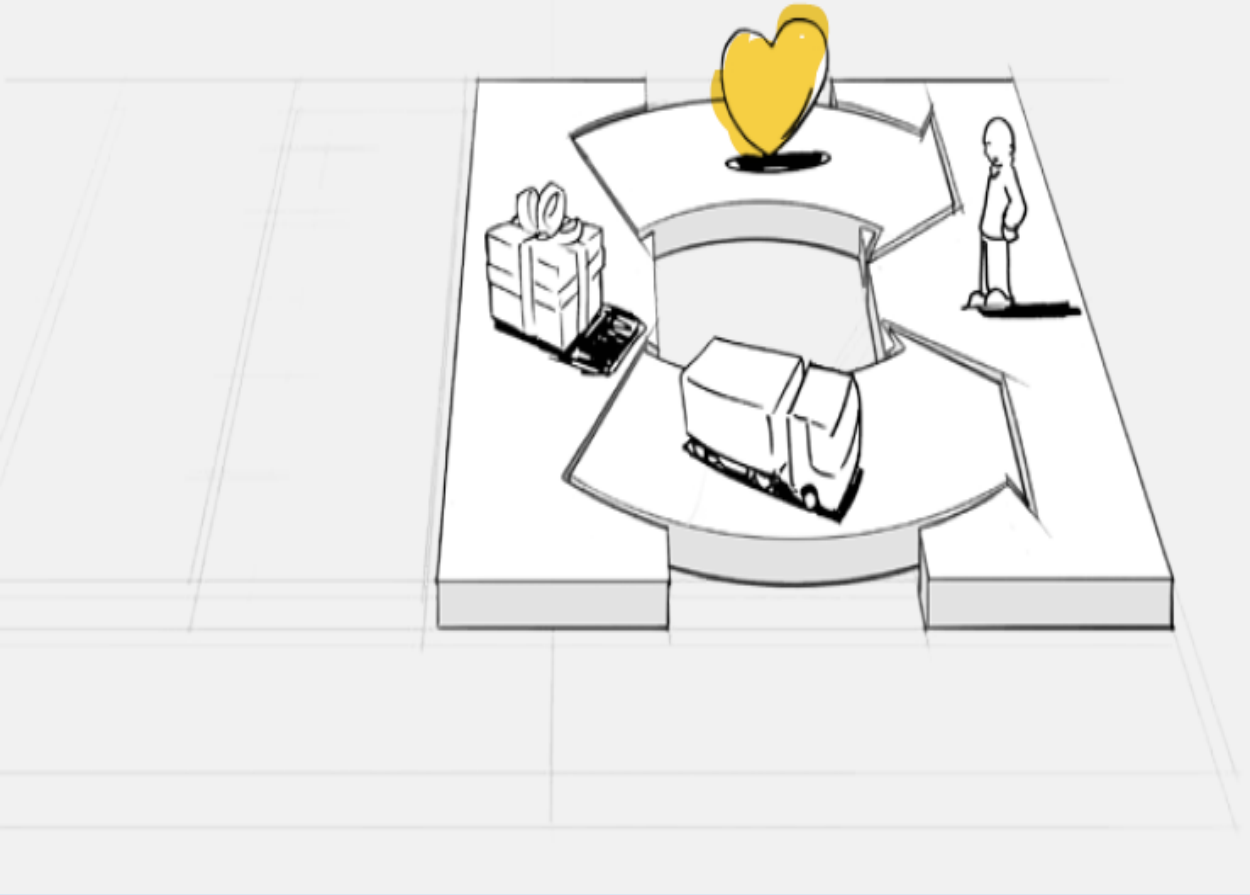
# Channels

**The Channels Building Block** describes how a company communicates with and reaches its Customer Segments to deliver a Value Proposition.



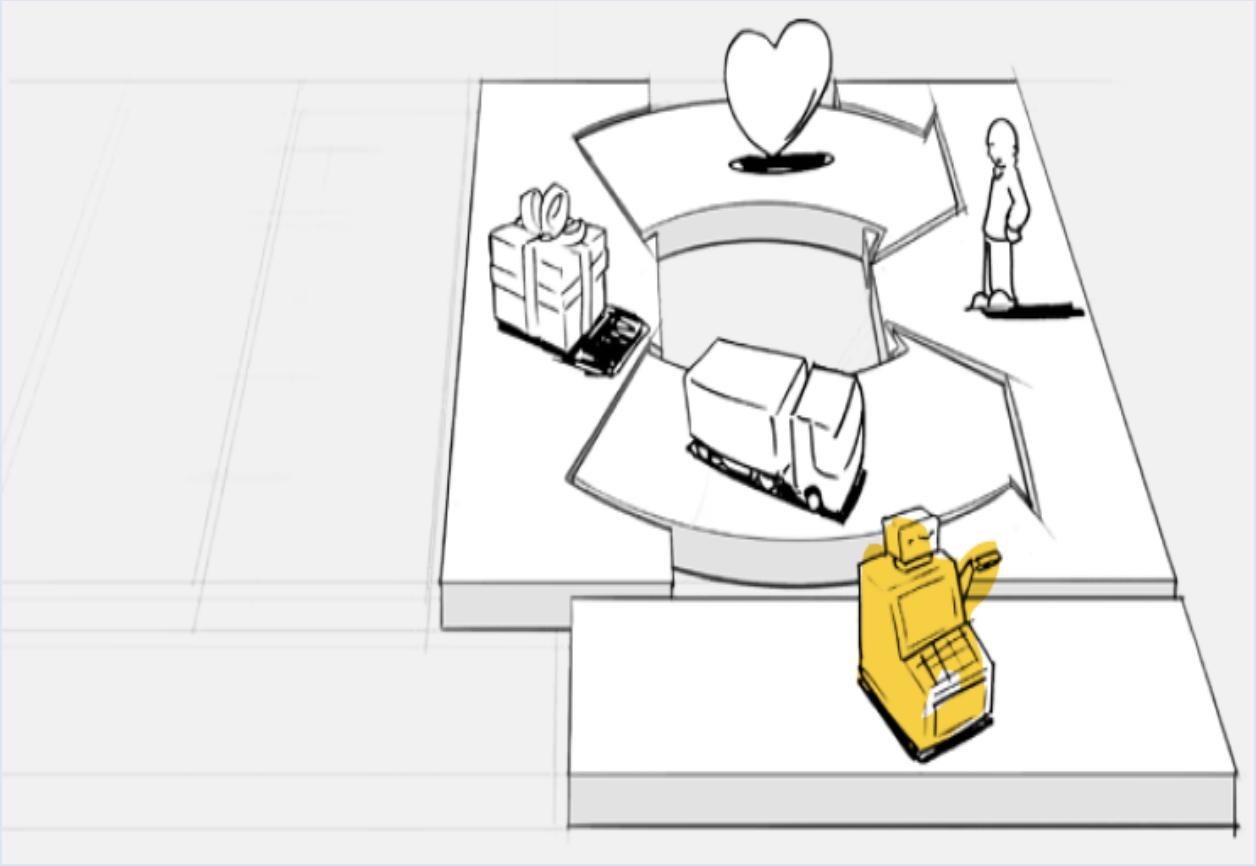
# Customer Relationships

**The Customer Relationships Building Block** describes the types of relationships a company establishes with specific Customer Segments



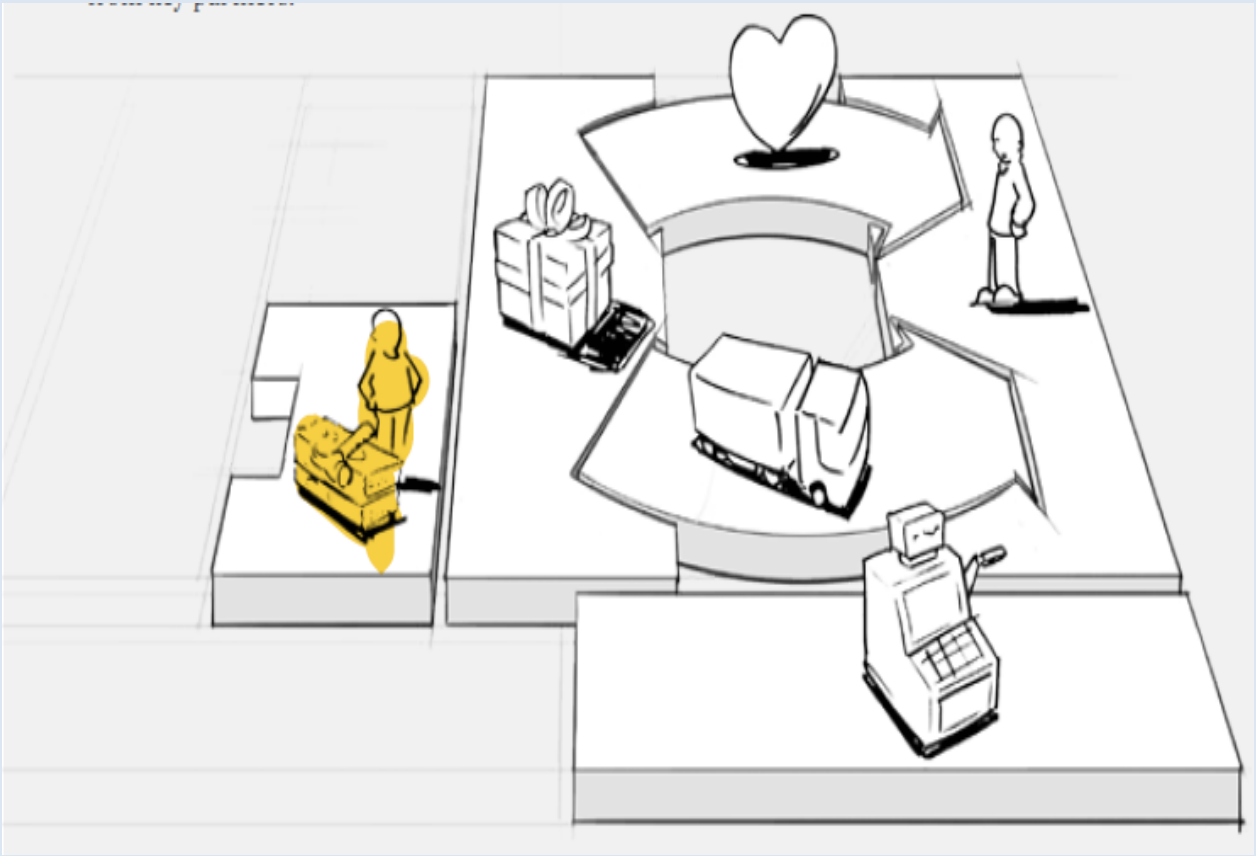
# Revenue Streams

**The Revenue Streams Building Block** represents the cash a company generates from each Customer Segment (costs must be subtracted from revenues to create earnings)



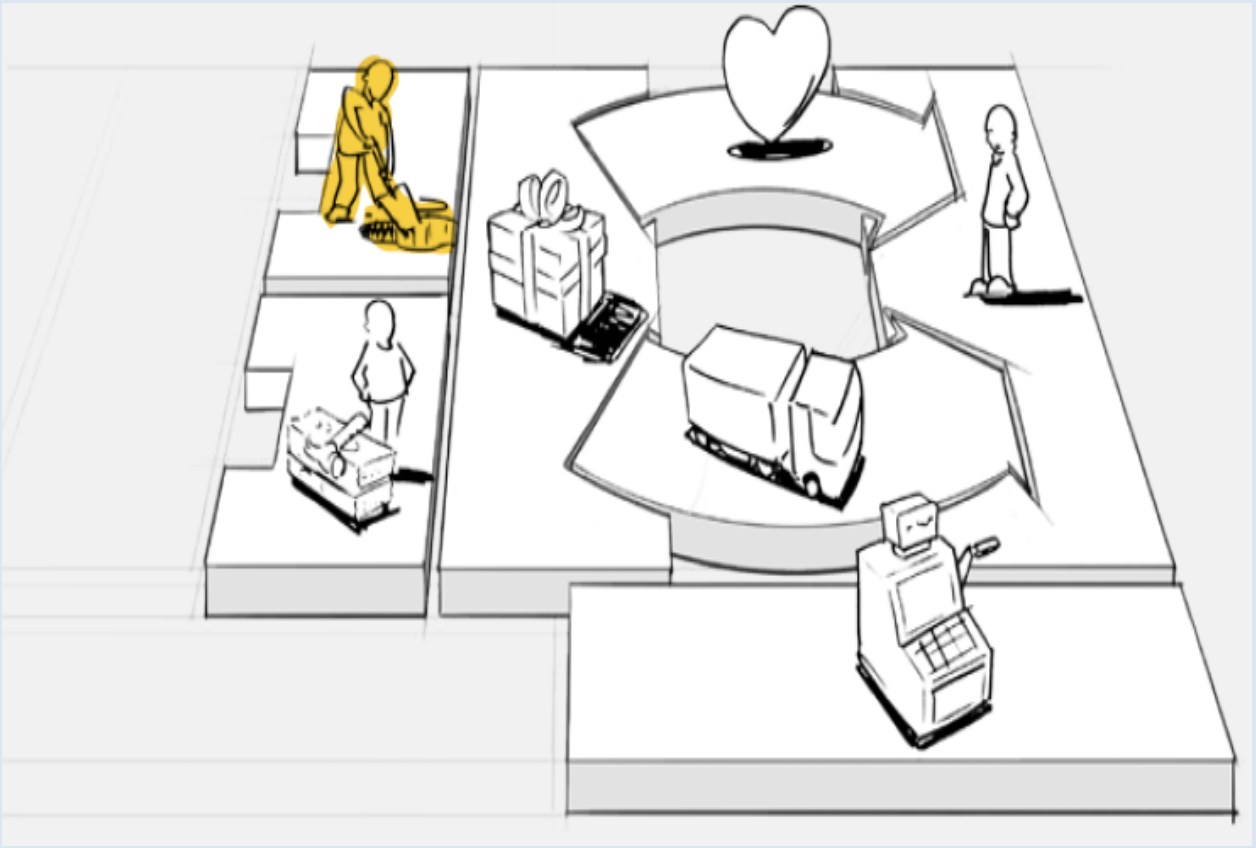
# Key Resources

**The Key Resources Building Block** describes the most important assets required to make a business model work



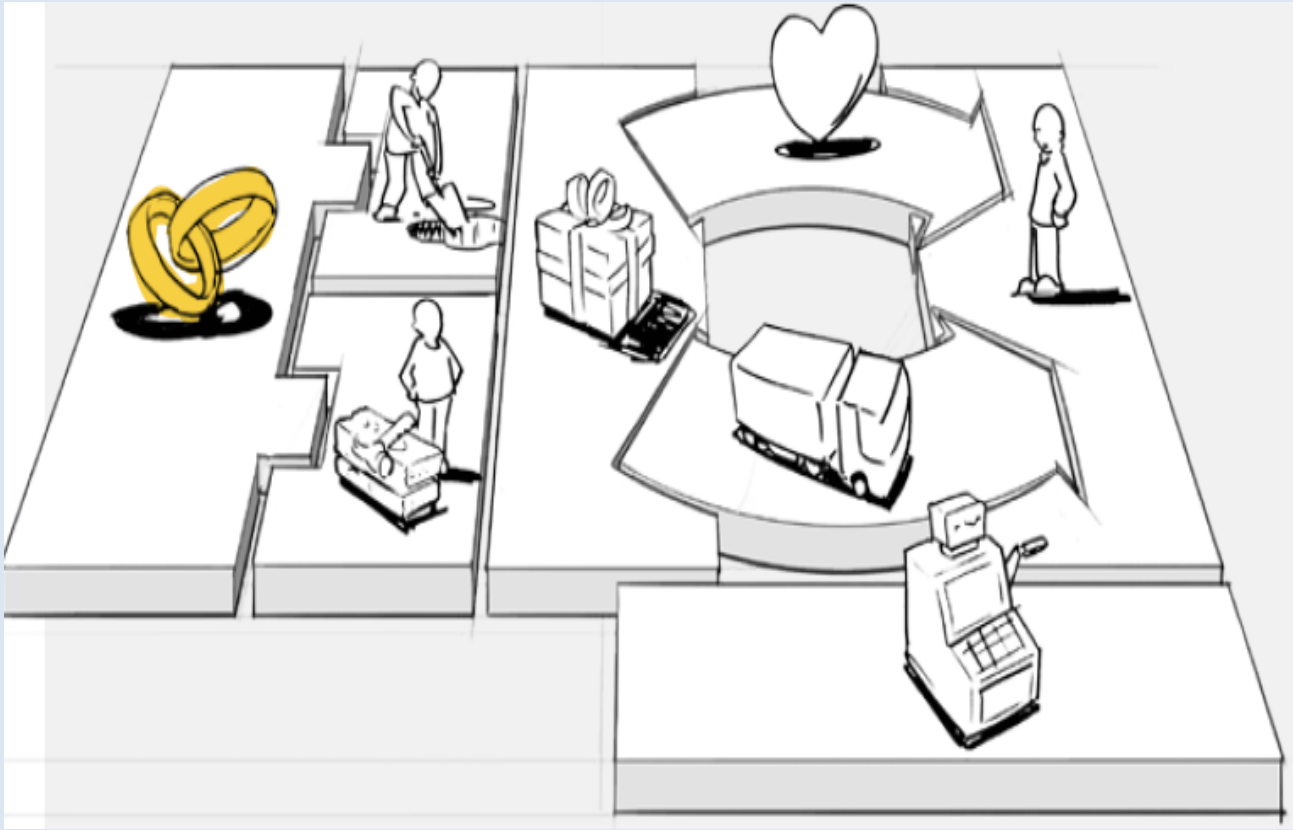
# Key Activities

**The Key Activities Building Block** describes the most important things a company must do to make its business model work



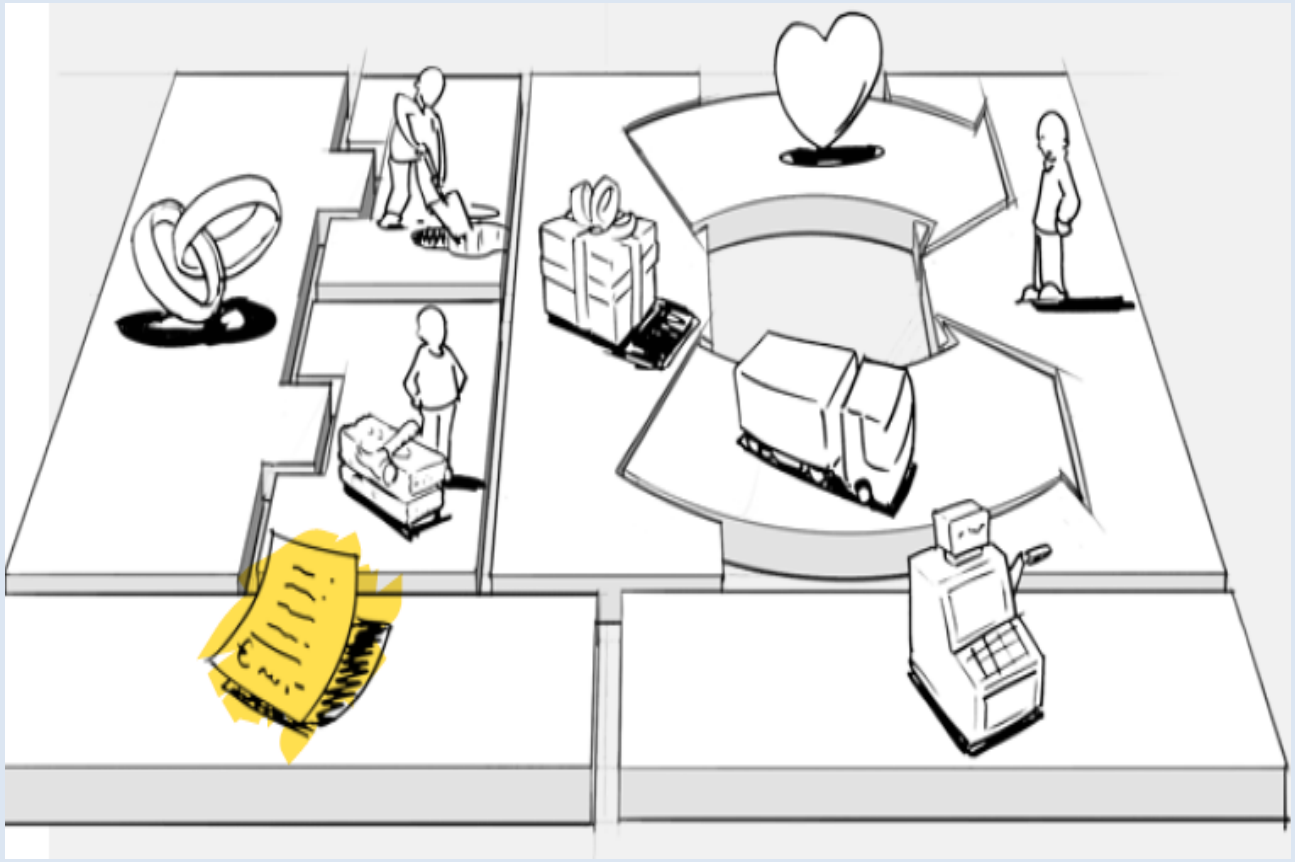
# Key Partnerships

**The Key Partnerships Building Block** describes the network of suppliers and partners that make the business model work



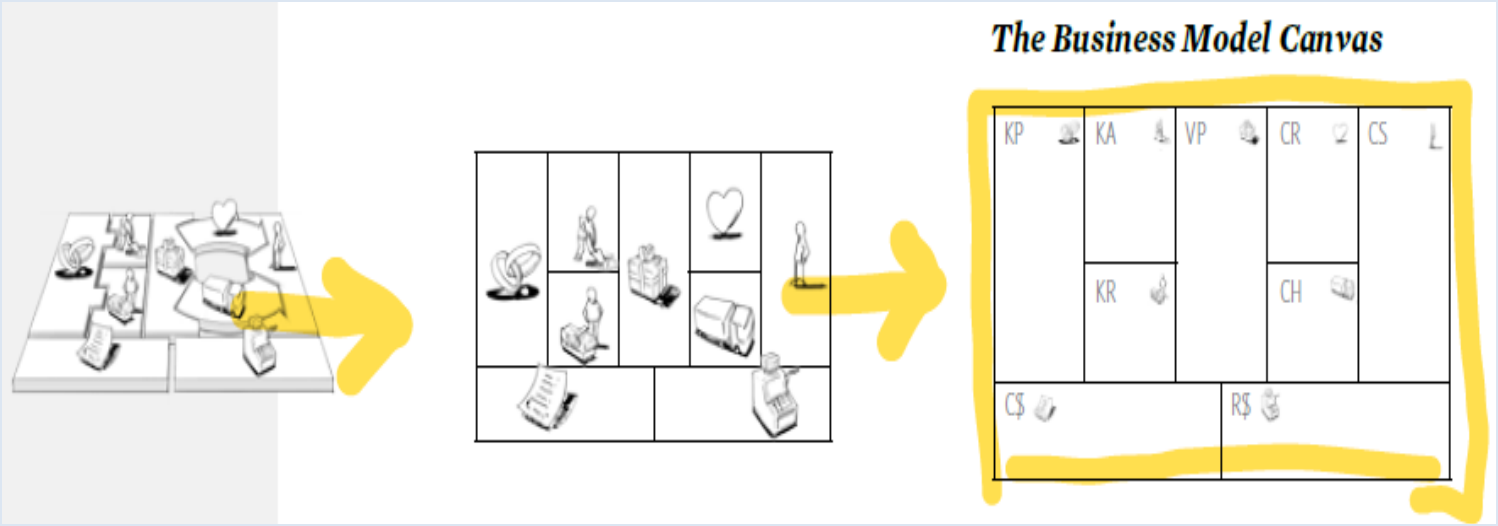
# Cost Structure

**The Cost Structure** describes all costs incurred to operate a business model



# The Business Model Canvas

The nine business model Building Blocks form the basis for a handy tool, which we call the ***Business Model Canvas.***




- This tool resembles a painter's canvas — preformatted with the nine blocks — which allows you to paint pictures of new or existing business models.
- ***The Business Model Canvas works best when printed out on a large surface so groups of people can jointly start sketching and discussing business model elements*** with Post-it notes or board markers. It is a hands-on tool that fosters understanding, discussion, creativity, and analysis.

# The Business Model Canvas



# IoT Business Models

- IoT based commercial applications are used in various fields such as manufacturing, health, energy, security, finance, etc., and IoT market has been rapidly growing each day.
- In the last 5 years, the number of devices used in IoT has increased 5 times and, in 2015, it is estimated as approx. 4.9 million.
- Cisco estimated that the business value of IoT application will be $14 trillion in the next decade.
- The business models and tools to create value for IoT Technologies and applications is a need.
- IoT business models is important to improve the success of IoT services.



**IoT business models aim to commercialize IoT applications. IoT creates big alterations in business model concepts.**

# What is the novelty added by IoT?

- The information resources become richer by the IoT elements consisted of smart devices such as smart phones, sensors, etc. The empirical data obtained by information sources have a significant effect to increase revenue.





General Information Sources









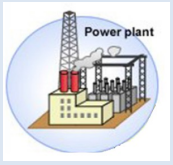

IoT-based information sources

- A service provider or a GSM operator can provide tailor-made solutions to their customers by real-time user statistics.

- A car insurance company can provide special offers to their customers by tracking and analyzing driver's behavior by vehicle sensor such as navigation, speed, line violation, accident detection, etc.



- Energy suppliers can also provide special offers to their customers by tracking and analyzing their energy consumption through IoT applications. Thus, it can optimize the energy consumption.



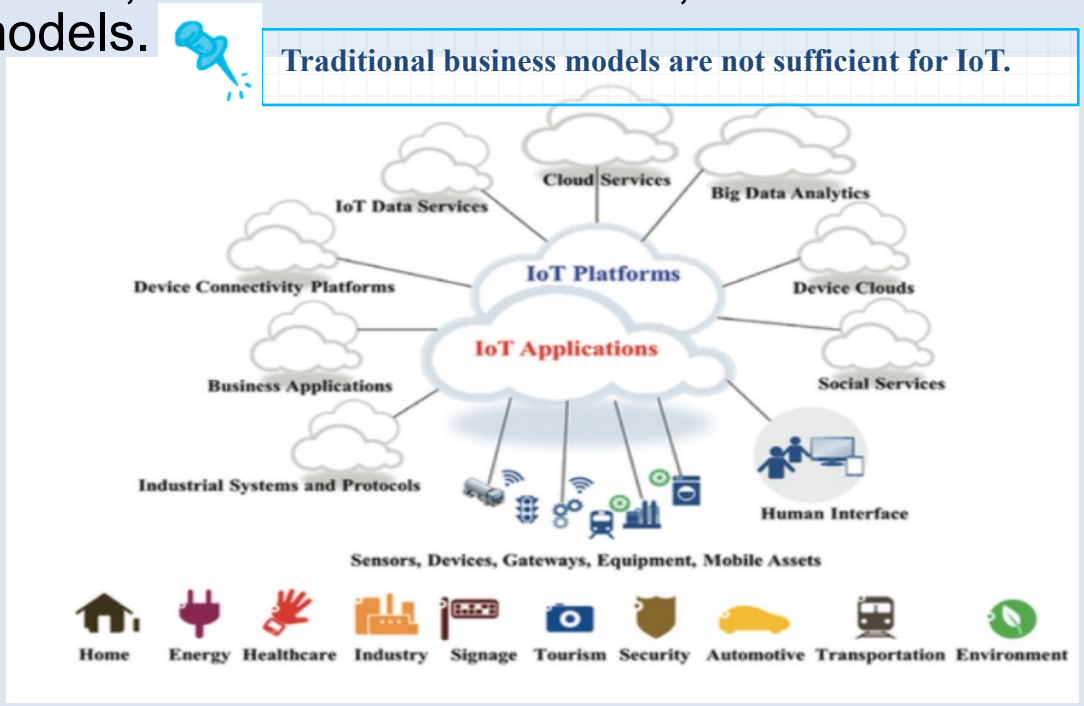






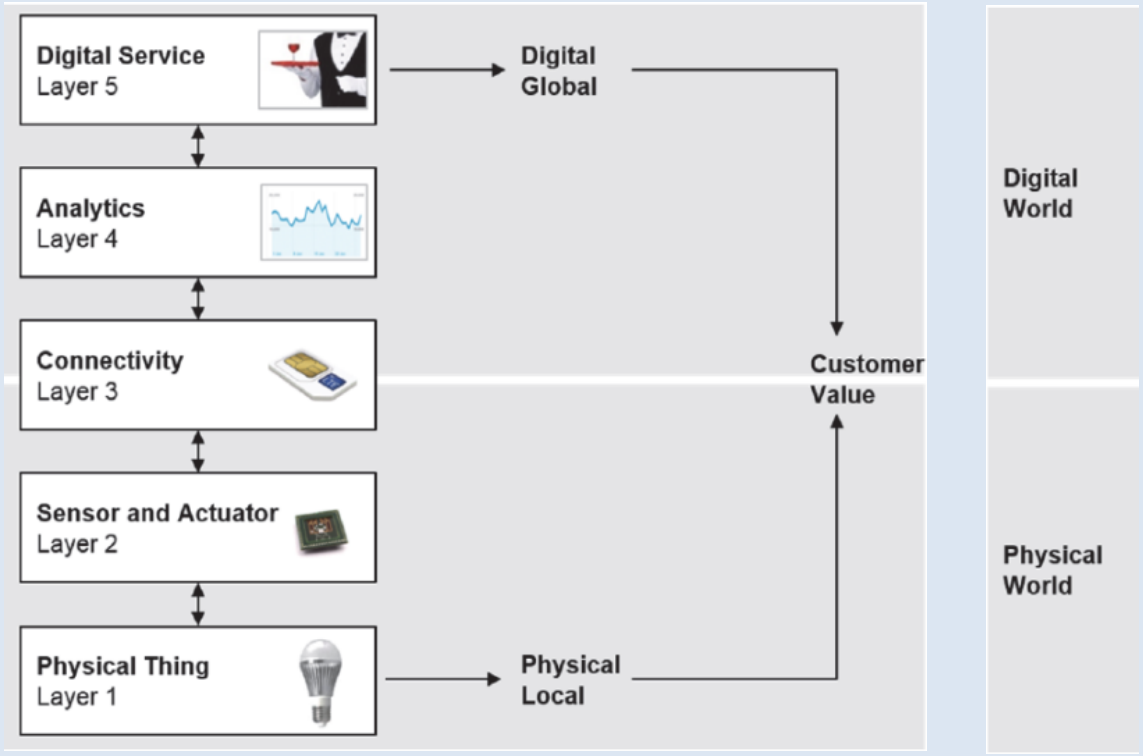
# The Benefits of IoT

- New Services,
- New business opportunities,
- Helps companies to create value.
- For instance, usage of mobile technology created new services such as mobile payment, mobile advertisement, location-based services, etc. into business models.



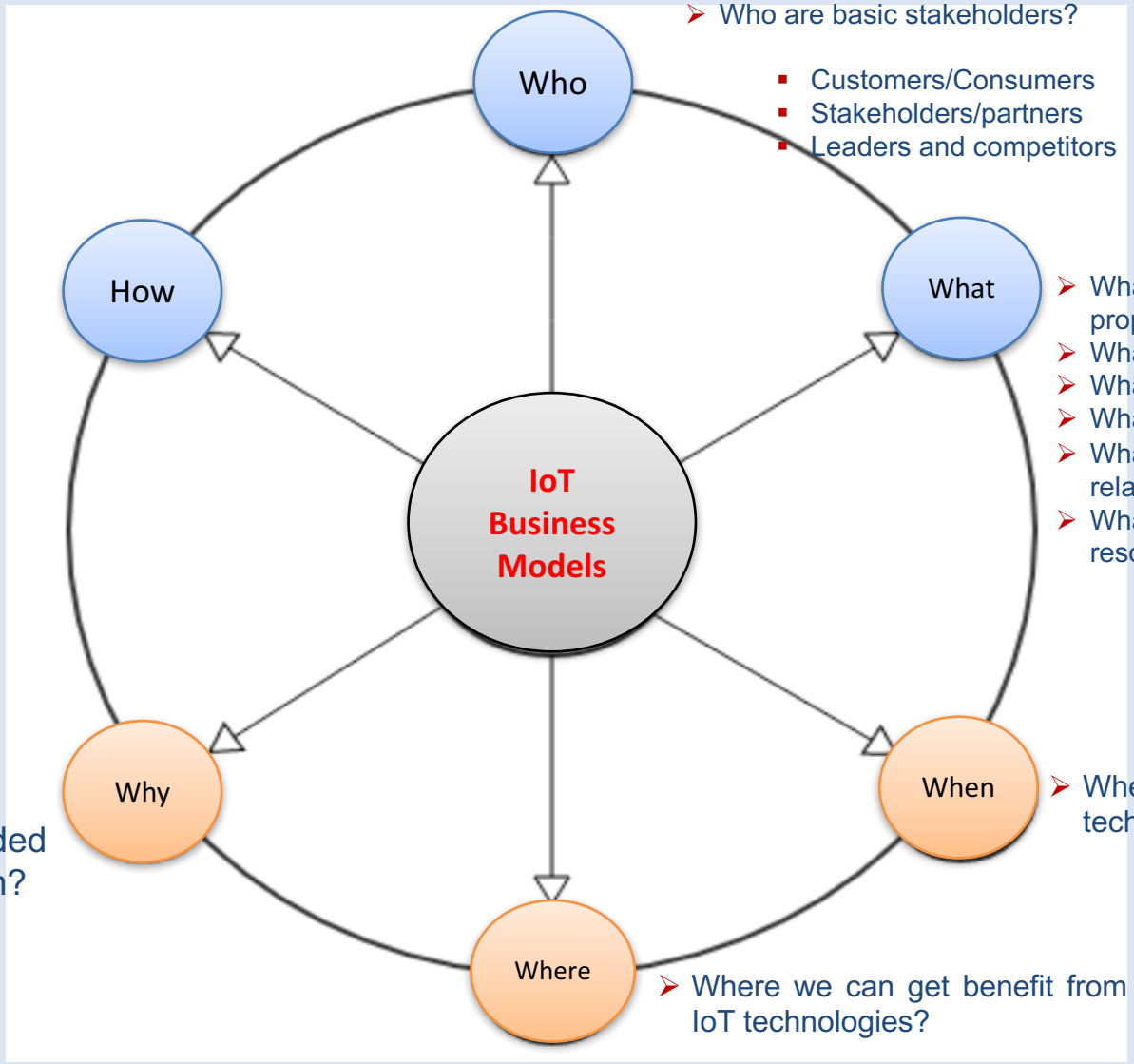
# The layers of creating value in IoT solutions

- Digital Services: Application layer The application (mobile, web, etc.) offered to customer.
- Analytics: Analysis of user data and creating user profile.
- Connectivity: Online Access to IoT hardware. Ex: Tracking location and movement of a bicycle
- Sensor and actuators: Microprocessor, sensor, and actuator. Runs locally. Data such as distance by accelerometer, location by GPS, battery state, etc.
- Physical Thing: The thing benefits to user. Bicycle, transportation, health activity.




Source: D. Bilgeri et al. ‘‘The IoT Business Model Builder ’’, White Paper, Bosch Software Innovations, GmbH. 2015.

# IoT Business Models R&D Criteria




- Who are basic stakeholders?
  - Customers/Consumers
  - Stakeholders/partners
  - Leaders and competitors

- How do IoT Technologies benefit companies/firms?
- How to create value through IoT value chain?

- What is value proposition?
- What is the cost?
- What is the benefit/profit?
- What are the channels?
- What are the customer relationships?
- What are the basic resources?

- Why are basic stakeholders added to IoT ecosystem?

- When to apply IoT technologies?
  - When the company big enough and ready ?

- Where we can get benefit from IoT technologies?

# IoT Business Model Questions

| Questions: | Organization | Industry | Ecosystem |
|---|---|---|---|
| **Who?** | - What skillset does the employees possess?<br>- Who are the company's main stakeholders? (customers, partners, suppliers, venders) | - Who are the main stakeholders?<br>- Key partners<br>- Competition<br>- Are there any levels of coopetition? | - Who is the ecosystem leader(s)?<br>- Who are potential new customers?<br>- Potential new partners<br>- Potential new competition/coopetition |
| **What?** | - What are the key channels?<br>- Key resources<br>- Cost structure<br>- Value proposition<br>- Value drivers<br>- Needs<br>- Links<br>- Value exchanges/ value extracts<br>- Flows<br>- core value<br>- What are other relevant activities?<br>- Risk<br>- State of the ecosystem | - What are the key channels?<br>- Key resources<br>- Cost structure<br>- Value proposition<br>- Value drivers<br>- Needs<br>- Links<br>- Value exchanges/ value extracts<br>- Flows<br>- core value<br>- What are other relevant activities?<br>- Risk<br>- State of the ecosystem | - What are the key channels?<br>- Key resources<br>- Cost structure<br>- Value proposition<br>- Value drivers<br>- Needs<br>- Links<br>- Value exchanges/ value extracts<br>- Flows<br>- core value<br>- What are other relevant activities?<br>- Risk<br>- State of the ecosystem |
| **How?** | - How is knowledge/ information exchanged throughout the company?<br>- How do the flows/ activities work?<br>- How does the system work?<br>- How is value created and captured?<br>- How is value exchanged/ extracted?<br>- How is the ecosystem structured/unstructured? | - How is knowledge/ information exchanged throughout the industry?<br>- How do the flows/ activities work?<br>- How does the system work?<br>- How is value created and captured?<br>- How is value exchanged/ extracted?<br>- How is the ecosystem structured/unstructured? | - How is knowledge/ information exchanged throughout the ecosystem?<br>- How do the flows/ activities work?<br>- How does the system work?<br>- How is value created and captured?<br>- How is value exchanged/ extracted?<br>- How is the ecosystem structured/unstructured? |

Kaynak: K. L. Fugl, "Business Model Framework Proposal for Internet of Things", Copenhagen Business School, 2015.

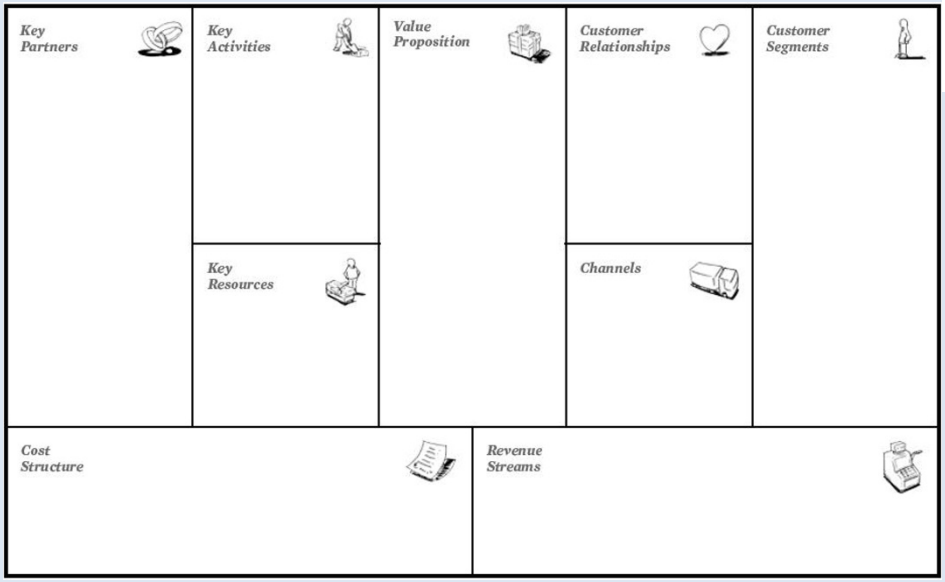
# IoT Business Model Questions

| Questions: | Organization | Industry | Ecosystem |
|---|---|---|---|
| **When?** | - When should the company evolve towards IoT? (after analyzing the who and what)<br>- When will the company's stakeholders adopt new innovation? | - When did/will the company's stakeholders move towards IoT?<br>- When will the company's stakeholders adopt new innovation? | - When should the company expand in the ecosystem? (move towards new markets)<br>- When will the company's stakeholders adopt new innovation? |
| **Where?** | - Where is the company in the IoT ecosystem?<br>- Where are the value creation/capture processes?<br>- Where are the value exchanges?<br>- Where are the information flows? | - Where is the industry in the IoT ecosystem?<br>- Where are the partners/competitors?<br>- Where are sources of opportunities in the industry?<br>- Where are the value creation/capture processes?<br>- Where are the value exchanges?<br>- Where are the information flows? | - Where is the IoT ecosystem the company is part of compared to the overall IoT environment?<br>- Where are the (potential) partners/competitors?<br>- Where are the potential customers?<br>- Where are the value creation/capture processes?<br>- Where are the value exchanges?<br>- Where are the information flows? |
| **Why?** | - What are the monetary/non-monetary benefits?<br>- Why should the company expand to new markets (if the when fits)? | - What benefits are involved for other actors in the industry?<br>- What are the monetary/non-monetary benefits? | - What benefits are involved for other actors in the ecosystem?<br>- What are the monetary/non-monetary benefits? |

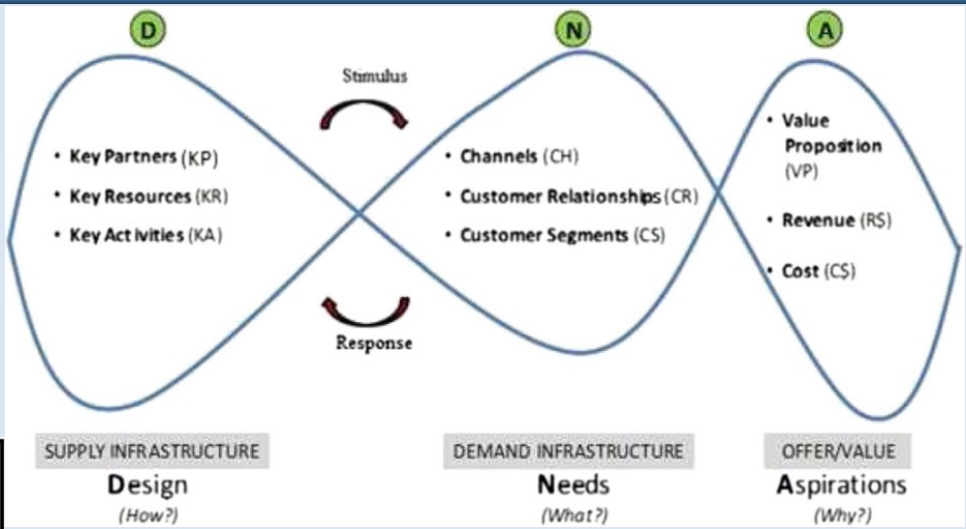
Kaynak: K. L. Fugl, "Business Model Framework Proposal for Internet of Things", Copenhagen Business School, 2015.

# Models proposed for IoT Business

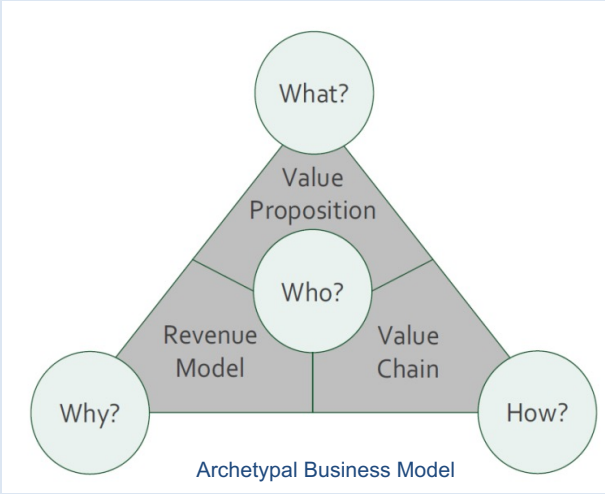
- Proposed business models for IoT
  - Business Model Canvas,
  - DNA Model,
  - Archetypal Business Model
  - Value Net Model,
  - MOP Model,




Business Model Canvas




DNA Model




Archetypal Business Model

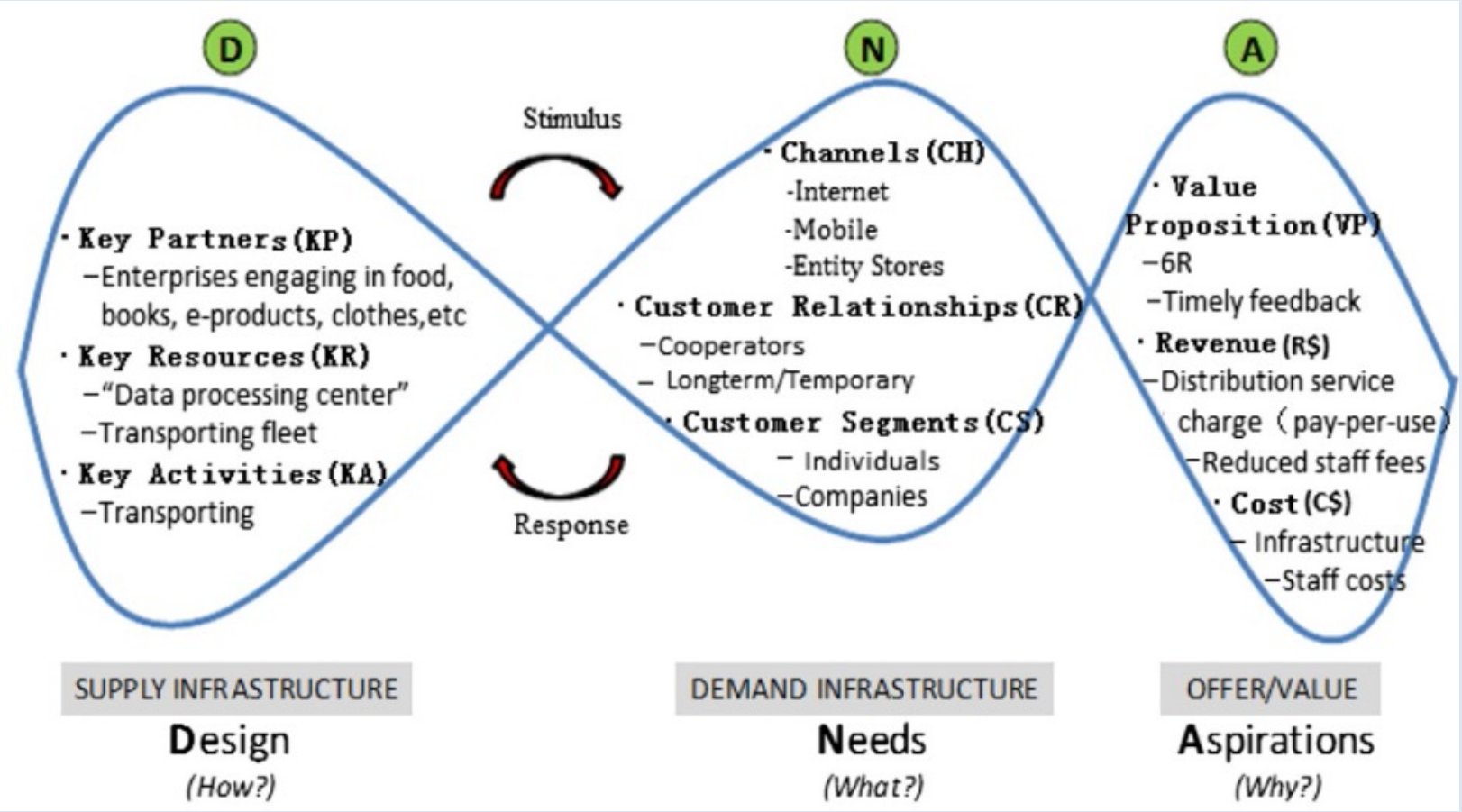
# Components in current IoT Business Models

| Main Perspectives | Components (Building blocks) | Key Elements |
|---|---|---|
| Infrastructure | Key partner | Software Developer, Data Analyst, Device Manufacturer |
| | Key resources | Software, Information, Customer Resource |
| | Key activities | Product Development, Platform Development, Partner Management, Platform & Resource Integration Ability |
| Value proposition | Value proposition | Convenience, Performance, Customization, Share |
| Customer | Customer relationship | Co-creation, Self-service Communication, Fast Feedback |
| | Customer segments | Mobile Users, Companies |
| | Channel | |
| Financials | Cost structure | IT Cost, Infrastructure |
| | Revenue structure | Subscription Fees, Usage Fee |

Source: J. Ju, M. Kim, J. Ahn, ‘‘Prototyping Business Models for IoT Service’’, Information Technology and Quantative Management (ITQM 2016), Procedia Computer Science, Vol. 91, 882-890, 2016

# Business Model Examples

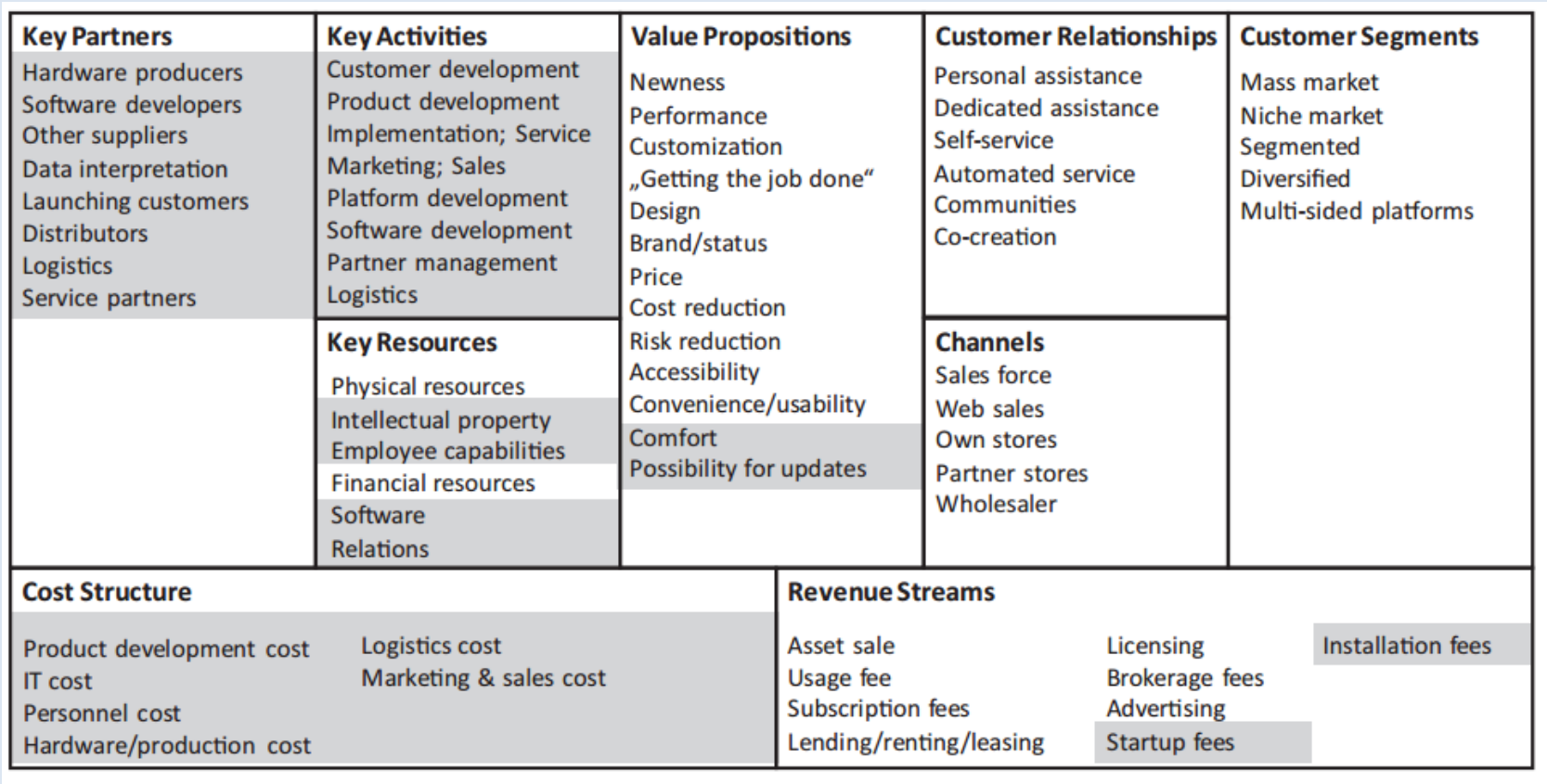
- **DNA-based business model for Smart Logistics**




Source: Y. Sun et al. ''A holistic approach to visualizing business models for the internet of things'', Communications in Mobile Computing, 1-7, 2012

# Business Model Examples

- **Canvas-based business model for IoT applications**




Source: R. M. Dijkman, B. Sprenkels, T. Peeters, A. Janssen, ‘‘Business models for the Internet of Things’’, International Journal of Information Management, Vol. 35, 672-678, 2015

# Business Model Examples

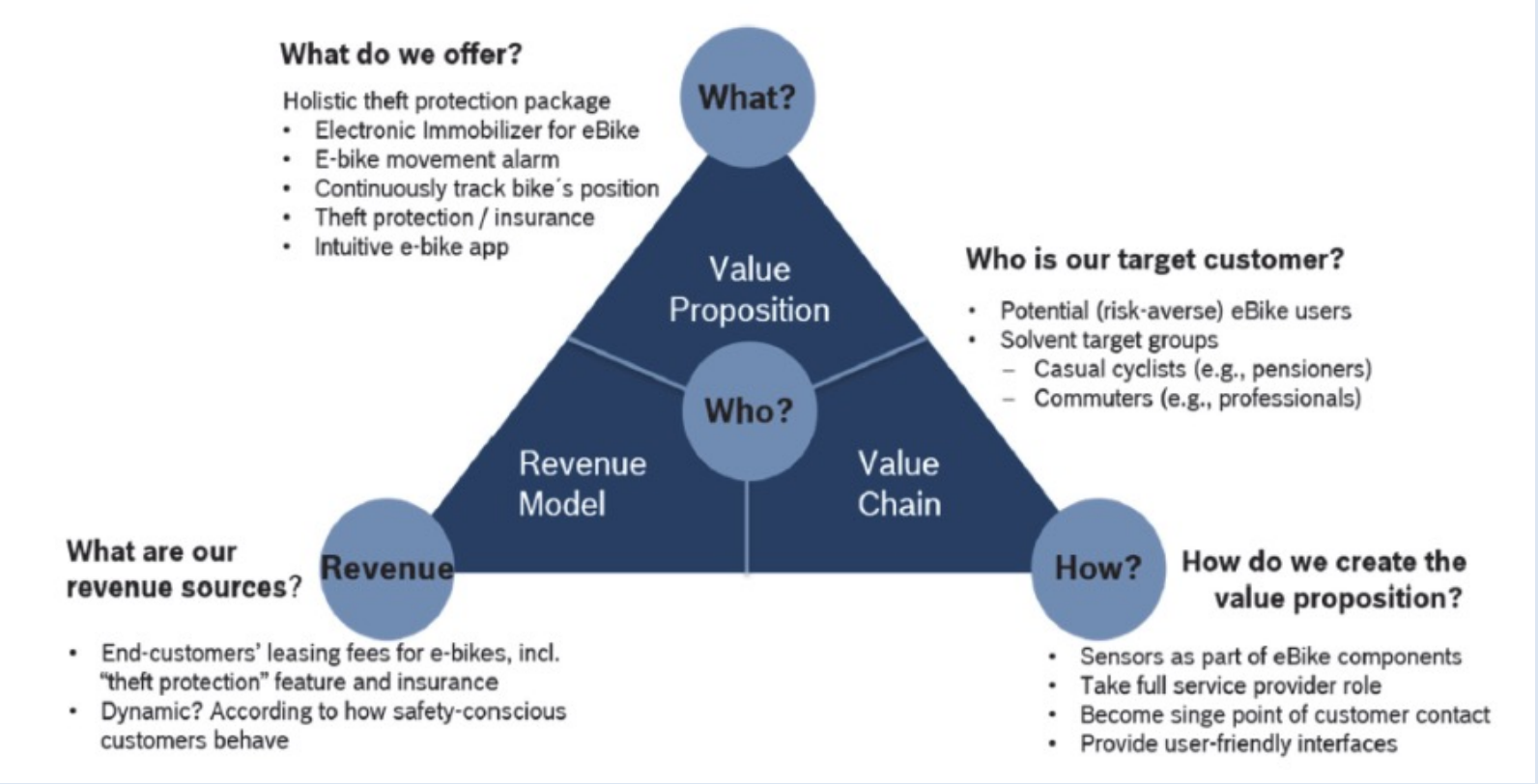
- **Canvas-based business model for IoT applications**

<table>
<tr><td>Key Partners</td><td>Key Activities</td><td rowspan="2">Value Propositions</td><td>Customer Relationships</td><td rowspan="2">Customer Segments</td></tr>
<tr><td rowspan="2">· Software Developer<br>· Data Analytics Company<br>· Device Manufacturer</td><td>· Product Development<br>· Partner Management<br>· Platform Integration</td><td>· Co-Creation</td></tr>
<tr><td>Key Resources</td><td>Channels</td><td>· General Customer Segment<br>· Vertical Market<br>· Global Market</td></tr>
<tr><td></td><td>· Sensors<br>· Cloud Service (Software)<br>· IoT Dedicated Network<br>· Capability for Business Analytics</td><td></td><td>· Internet<br>· Mobile</td><td></td></tr>
<tr><td colspan="3">Cost Structure</td><td colspan="2">Revenue Streams</td></tr>
<tr><td colspan="3">· IT Cost<br>· Maintenance</td><td colspan="2">· Profit sharing<br>· Subscription fee<br>· Product sales</td></tr>
</table>

Source: J. Ju, M. Kim, J. Ahn, ‘‘Prototyping Business Models for IoT Service’’, Information Technology and Quantative Management (ITQM 2016), Procedia Computer Science, Vol. 91, 882-890, 2016

# Business Model Examples

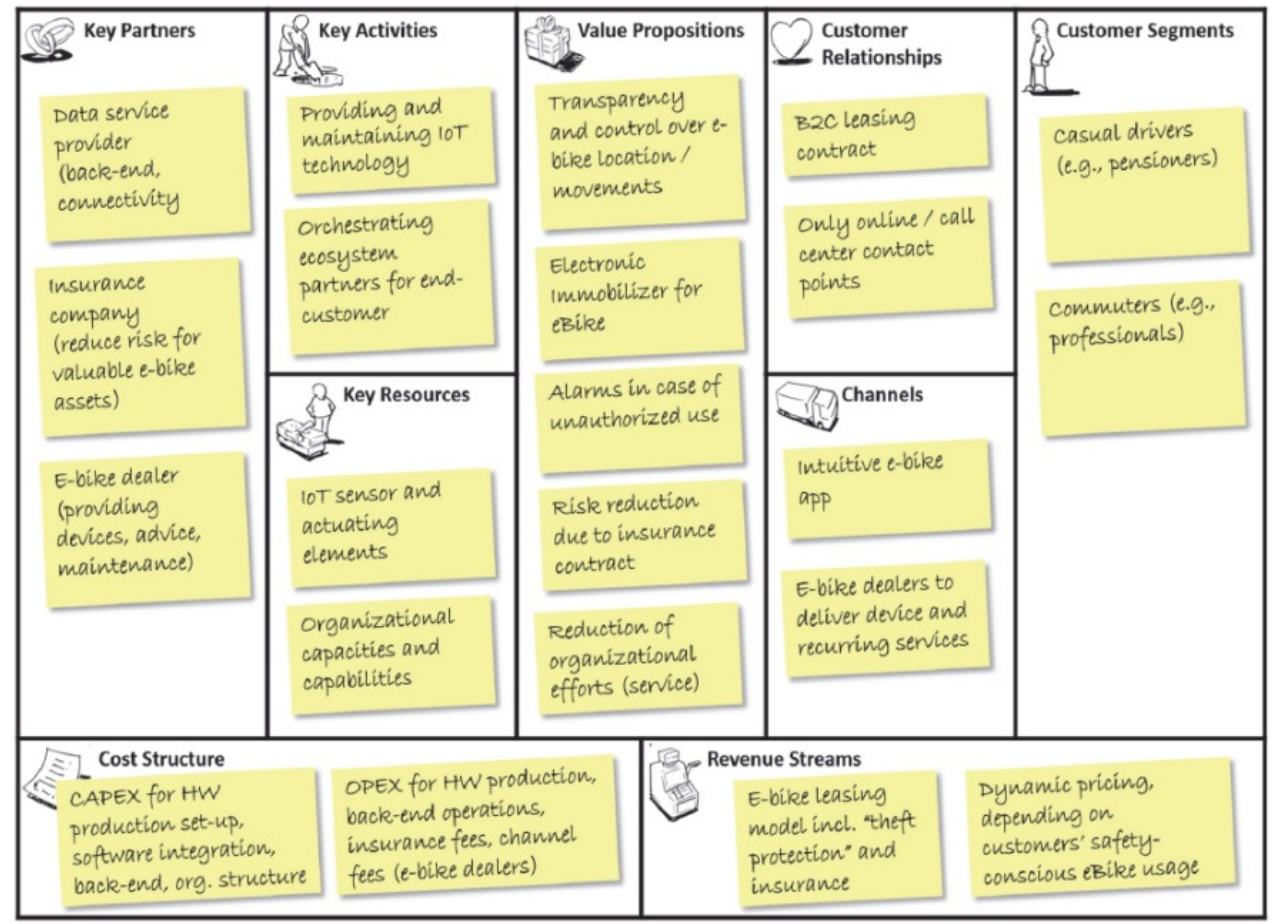
- **Archetypal (Galler Magic Triangle)-based business model for e-bike**




Source: D. Bilgeri et al. ‘‘The IoT Business Model Builder ’’, White Paper, Bosch Software Innovations, GmbH. 2015.

# Business Model Examples

- **Canvas-based business model for e-bike**




Source: D. Bilgeri et al. ''The IoT Business Model Builder '', White Paper, Bosch Software Innovations, GmbH. 2015.

# IoT Business Model for Different Fields

| Building Blocks | Elements | Google (Smart Home) | GE (Industrial IoT) | Car2Go (Transportation) |
|---|---|---|---|---|
| Key Partners | Software Developer | In-house development | In-house development | Outsourcing |
| | Data Analyst | In-house analysis | In-house analysis | In-house analysis |
| | Device Manufacturer | In-house development | In-house development | Manufacturer for maintenance and repairs |
| Key Resources | Sensors | Sensors | Sensors | Sensors |
| | Cloud Service (Software) | Mobile application<br>Analysis software | Predix Cloud (Software platform) | Mobile application<br>Analysis software |
| | Capability for Business Analytics | In-house analyst | In-house analyst | In-house analyst |
| Key Activities | Product development | Thermostat (3rd generation) | All industrial parts equipped with sensors | Fleet management with sensors (e.g., optimization) |
| | Partner management | Partnership with other IoT services (e.g., light bulbs) | IoT alliance (e.g., Intel, Cisco) | - |
| | Platform integration | 'Works with Nest' integration | Industrial Cloud-based platform | Transportation platform (city transit services) |
| Value Proposition | Performance | Efficient operation | Predictive maintenance efficiency | Intelligent mobility for customers |
| | Convenience | Automation | Real-time data tracking | Flexible urban mobility |
| | Customization | Customized schedule | Customized total solutions | On-demand access |

Source J. Ju, M. Kim, J. Ahn, "Prototyping Business Models for IoT Service", Information Technology and Quantative Management (ITQM 2016), Procedia Computer Science, Vol. 91, 882-890, 2016

# IoT and Traditional Business Models

| | Traditional product | IoT |
|---|---|---|
| Customer Needs | Solutions for current needs | Current and possible needs in a foreseeable future |
| Proposition | Independent product that becomes out-of-date in time | Update remotely |
| Role of Data | Single-point data for future product requirements | Services for data transformation and provides experience for future products |
| Profit (Benefit) | Next product or device sale | Continuous revenue |
| Control | Commercial product advantages | Personalized products |
| Skill improvement | Current resources | It understands how other partners and competitors earn money |

# References


- A. Osterwalder, Y. Pigneur, "Business Model Generation", Çev. Melis İnan, Optimist Yayım Dağıtım, 2013, http://www.businessmodelgeneration.com/
- R. M. Dijkman, B. Sprenkels, T. Peeters, A. Janssen, "Business models for the Internet of Things", International Journal of Information Management, Vol. 35, 672-678, 2015
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